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End Semester Examination 2025

Name of the Course: B. Tech
 Name of the Paper – Power System Protection
 Paper Code: TEE 603

Semester: VI

Time: 3Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1														
(a)	Classify the different faults in power system. Which of these are more frequent?															
(b)	Differentiate between (i) Switch and Switchgear (ii) Fuse and Circuit Breaker															
(c)	Why protection system is provided in power system? Why overlapping of protective zones desirable?															
Q2	(20 marks)	CO2														
(a)	Define PSM. Sometimes there is a need to replace the old current transformers with new ones for upgrading the system. In this case, it is required to change the value of the Plug Setting Multiplier, and no need to change the TMS. While changing Old CT with New CT we must keep in mind that the relay pick-up current should not be changed. Based on the following information find out the new PSM value. Old CT Ratio- 75/1 A, PSM – 0.75 New CT Ratio- 100/5A															
(b)	An overcurrent relay of 5A is used to protect a feeder through 1000/5 A CT. the relay has a PS of 50% and TMS = 0.8 . Find the operation of the said relay if a fault current of 1200 A flows through the feeder. Make use of the following characteristic <table><tr><td>PSM</td><td>1.3</td><td>2</td><td>4</td><td>6</td><td>10</td><td>20</td></tr><tr><td>Time for unity TMS (sec)</td><td>30</td><td>10</td><td>6.5</td><td>3.5</td><td>3</td><td>2.2</td></tr></table>		PSM	1.3	2	4	6	10	20	Time for unity TMS (sec)	30	10	6.5	3.5	3	2.2
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(c)	Discuss the working and construction of an induction type overcurrent relay.															

Q3	(20 marks)	CO3
(a)	Explain Merz Price system of protection as applied to alternator.	
(b)	Discuss a protection scheme employed for the protection of power transformers in case of an internal fault.	
(c)	What is an impedance relay? Explain its operating principle, torque equation and operating characteristics.	
Q4	(20 marks)	CO4,5
(a)	List various building blocks that constitute the static relay and give brief description about each of the blocks.	
(b)	Explain how arc is initiated and sustained in a circuit breaker when the circuit breaker contacts separate.	
(c)	In a short circuit test on a 110 kV, 3-phase system, the breaker gave the following results: pf. of the fault 0.4, recovery voltage 0.95 of full line value; the breaking current is symmetrical and the restriking transient had a natural frequency of 15 kHz. Determine the average rate of rise of restriking voltage. Assume that the neutral is grounded and fault involves earth.	
Q5	(20 marks)	CO5, CO6
(a)	Explain the terms (i) restriking voltage; (ii) recovery voltage; and (iii) RRRV.	
(b)	A 33 kV, 500MVA circuit breaker suddenly closes on to a fault. Determine (i) Symmetrical Breaking Current; (ii) asymmetrical breaking current assuming 50% dc component (iii) peak making current.	
(c)	Discuss the operating principle of SF6 circuit breaker, what are its advantages over other types of circuit breakers and for what voltage range it is recommended.	